

Abigail Lin

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Education

University of Florida | Bachelor of Science in Computer Science | Math Minor

August 2022 - May 2026

GPA: 3.82/4.00

Relevant Coursework: Linear Algebra for Data Science, Intro to Numerical Analysis, Algorithm Abstraction and Design, Introduction to Probability, Mathematics for Intelligent Systems (Fall 2025)

Research Experience

Purdue University, RCAC Anvil REU

May 2025 - August 2025 | Rosen Center For Advanced Computing

- Designed and implemented a bioinformatics workflow template generator (Bash, Slurm, Python, Urwid) providing a simpler, more accessible alternative to Nextflow and Snakemake.
- Used Apptainer and Docker for container creation and deployment to efficiently organize and manage software dependencies.

University of Florida, AI Scholars Program

January 2024 - May 2025 | Voiniciuc Lab | Xie Lab

- Generated 30 CSLA enzyme variants using a Maximum Entropy (MaxEnt) probabilistic AI model on UF's HiPerGator supercomputer.
- Automated experimental assays to evaluate the efficiency of generated enzyme variants in synthesizing β -mannans.
- Benchmarked ProteinGym mutation scores using generative protein language models, including ESM2 and SaProt.

University of Central Florida, Applied Computational Mathematics NSF REU

May 2024 - July 2024 | Dr. Carlos Borges

- Worked for 10 weeks as an NSF-funded summer undergraduate researcher.
- Developed a MATLAB solver for the forward scattering problem of objects with thin coatings, comparing impedance and transmission approximations and achieving a final error below 10^{-2} .

University of Florida, Research Assistant

January 2024 - Present | Voiniciuc Lab

- Developed Python programs to automate synthetic biology workflows using four Opentrons OT-2 lab robots, reducing pipetting error and increasing experimental efficiency.
- Wrote Python scripts to automate batch editing of GenBank files in Benchling, leveraging Biopython and Pandas for data parsing and processing.
- Conducting research on generative AI for plant enzyme engineering, applying AI models to design and validate enhanced β -mannanase (MAN) enzymes.

Publications	Qiande, M., Lin, A., Larson, L., & Voiniciuc, C. (2025). BOTany Methods: Accessible Automation for Plant Synthetic Biology. <i>bioRxiv</i>, 2025-08. (Under revision as a “Breakthrough Technology” article for <i>Plant Physiology</i>).
Presentations	Summer Undergraduate Research Symposium July 2025 Purdue University • “Empowering Research: Easy-to-Use Bioinformatics Workflow Templates,” <i>Undergraduate Research Symposium</i>, Purdue University, West Lafayette, Indiana, July 2025. 2025 UF Spring Undergraduate Research Symposium April 2025 University of Florida • “Using the Maximum Entropy Model to Enhance CSLA Enzyme β-Mannan Polysaccharide Synthesis,” <i>2025 UF Spring Undergraduate Research Symposium</i>, University of Florida, Gainesville, Florida, April 2025. Summer Poster Showcase July 2024 University of Central Florida • “Forward Scattering for Thin Coated Domains: Numerical Solution Using Generalized Impedance Boundary Conditions,” <i>Summer Poster Showcase</i>, University of Central Florida, Orlando, Florida, July 2024.
Involvement	Phi Sigma Rho January 2024 - Present • Participated in various sorority events, including volunteering, fostering a supportive community for women in STEM fields. Women in Computer Science and Engineering (WiCSE) September 2023 - December 2023 • Mentored a computer science major underclassman, provided guidance and technical instruction as well as, fostering their academic and professional growth.